



1502 - A Comparison of Abrasion Resistance Using Polyfluo® 190 and Fluo 300

Introduction

Polyfluo 190 and Fluo 300 are both designed to offer superior abrasion resistance. Polyfluo 190 is a unique PTFE- modified polyethylene. Fluo 300 is micronized polytetrafluoroethylene (PTFE).

	Polyfluo	Fluo 300
Maximum Particle Size	31.0	22.0
Mean Particle Size (µm)	9.0-12.0	5.0-6.0
Melting Point (°C)	121-132	>316
Density @ 25°C (g/cc)	0.99	2.2
NPIRI Grind	4.0-6.0	1.0-2.0

Objective

Compare the level of abrasion resistance between Polyfluo 190 and Fluo 300.

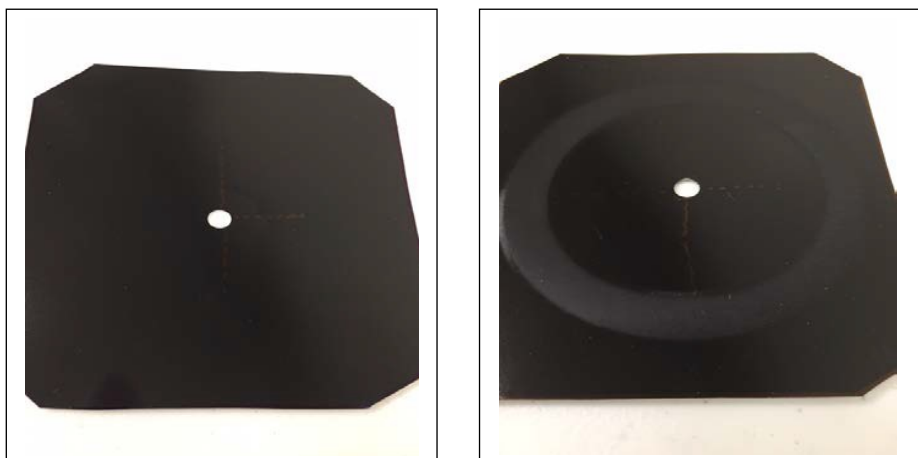
Study Methodology

Aqueous polyurethane dispersions were made at a 0.5%, 1.0% and 1.5% dosages (based on total formula weight) for each product. The dispersions were drawn down using a 0.003" (76 µm) Bird applicator on a coated lacquered Leneta Chart Form 2A. Panels were dried for one week before measurements were taken.

A Taber Abraser was used following ASTM D4060. The weight loss was measured in milligrams and recorded after 1000 cycles using 1000 gram auxiliary masses on each CS-17 wheel.

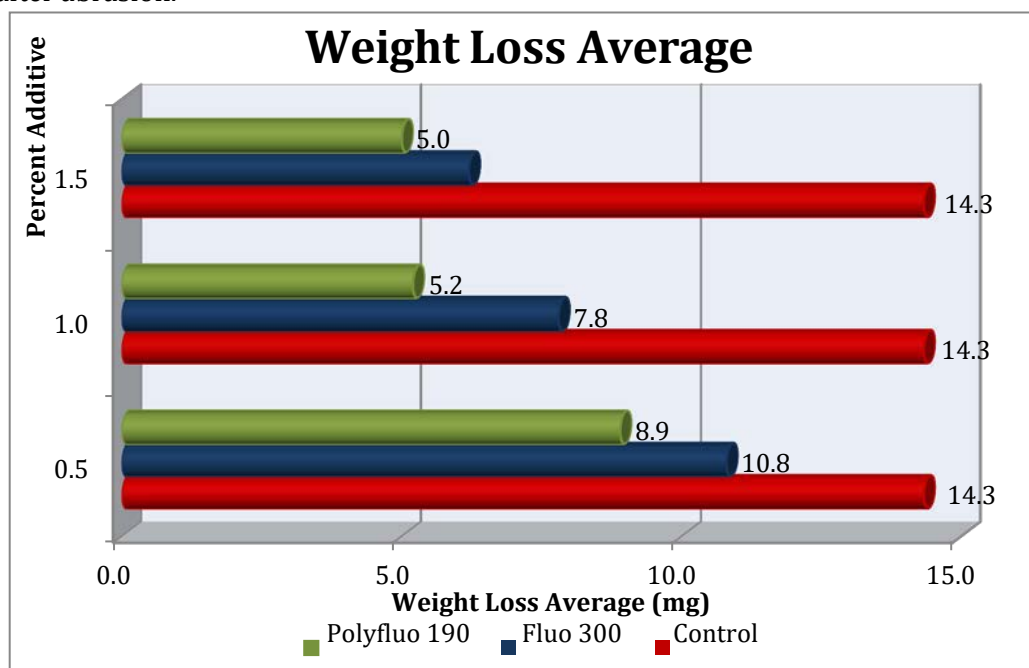


Test Panels Before and After



Results

The data below represents the average of 6 results measured by the weight loss equation $L = A - B$, where L equals the weight loss, A equals the weight of the specimen before abrasion and B equals the weight of the specimen after abrasion.



Conclusions

- Polyfluo 190 and Fluo 300 provide a similar level of Taber abrasion resistance
- Polyfluo 190 (PTFE-modified polyethylene) is slightly superior to Fluo 300 (100% PTFE)
 - Optimum Polyfluo 190 dosage level is around 1%
- Polyfluo 190 gives better abrasion resistance at lower cost
 - Polyfluo 190 is lower in cost than 100% PTFE additives

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